

Puru Asthana

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SUMMARY

AI/ML/Data Science enthusiast and B.Tech CSE student with hands-on experience in machine learning, NLP, deep learning, and EEG signal processing. Skilled in building end-to-end ML pipelines, predictive modeling, and feature engineering. Seeking AI/ML/Data Science roles to apply expertise in statistical analysis and model development.

EDUCATION

B.Tech in Computer Science Engineering

2023 - Present

Jaypee University of Engineering and Technology, Guna, India

CGPA: 7.0

- **Relevant Coursework:** Machine Learning, Data Structures and Algorithms, Statistics and Probability, DBMS, Linear Algebra, Artificial Intelligence

SKILL SUMMARY

Languages: Python, C/C++, SQL, HTML/CSS

ML/DL Frameworks and Libraries: Scikit-learn, PyTorch (basic), SpaCy, Hugging Face (basic), Pandas, NumPy, Matplotlib, Seaborn, SciPy, Streamlit

ML Concepts: Supervised Learning, Unsupervised Learning, Classification, Regression, Clustering, Ensemble Methods, Neural Networks, NLP, Hyperparameter Tuning, Cross-Validation

Tools and Platforms: Git, GitHub, VS Code, Google Colab, Jupyter Notebook, Microsoft Excel

EXPERIENCE

Maulana Azad National Institute of Technology (MANIT), Bhopal

On-site

Research Intern

June 2025 - July 2025

- Designed an end-to-end EEG emotion classification pipeline using the DREAMER dataset with preprocessing (filtering, normalization, noise handling) for feature extraction and ML model input.
- Built and evaluated MLP, Random Forest, and KNN models, then combined into a hybrid ensemble achieving **72% accuracy** on valence-arousal prediction, improving robustness over single-model baselines.

PROJECTS

Emotion Detection Using EEG Signals — Python, Scikit-learn, Ensemble ML, Signal Processing (GitHub)

- Built an EEG-based emotion classifier on DREAMER and DEAP datasets, predicting valence-arousal states with **72% accuracy** using a hybrid ensemble model (MLP, Random Forest, KNN).
- Implemented full signal preprocessing (filtering, normalization, noise handling) and feature extraction pipeline to maximize model input quality.

Named Entity Recognition (NER) — NLP, Deep Learning, Python, spaCy, Streamlit (GitHub)

- Trained a custom spaCy NER model to classify entities (persons, organizations, locations) from raw text; built full NLP pipeline (preprocessing, training, hyperparameter tuning, evaluation) and deployed via Streamlit.

Movie Recommendation System — Python, Scikit-learn, Machine Learning, Streamlit (GitHub)

- Built a content-based recommendation engine using TF-IDF vectorization and cosine similarity on movie metadata (genres, cast, directors, plot keywords), deployed as an interactive Streamlit web app.

CERTIFICATIONS

Machine Learning Specialization (Certificate)

April 2025

Stanford University and DeepLearning.AI via Coursera

- Supervised Learning (Regression, Classification, SVMs), Neural Networks and TensorFlow, Unsupervised Learning, Recommender Systems and Reinforcement Learning

POSITIONS OF RESPONSIBILITY

Vice Chairperson, Geeks for Geeks Student Chapter

2025 - Present

Jaypee University of Engineering and Technology

- Organizing technical workshops, coding events, and knowledge-sharing sessions for 50+ student members to foster a data science and programming culture.

Co-Head, Hospitality Team, TEDxJUET

2024

Jaypee University of Engineering and Technology

- Led the hospitality operations for TEDxJUET, coordinating end-to-end speaker management for 12 speakers including scheduling, accommodation, event-day needs, and on-ground support throughout the event contributing to a successful independently organized TEDx event.